Building Outline Delineation From VHR Remote Sensing Images Using the Convolutional Recurrent Neural Network Embedded With Line Segment Information

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***Abstract*— Recently, several recurrent neural network (RNN)-based models have been proposed to delineate the outlines of buildings from very high resolution (VHR) remote sensing images. These models first use convolutional neural networks (CNNs) to recognize the boundary fragments by learning probability maps of both edges and corners and then feed them into RNN to find and link a set of sequent corners into external boundaries of buildings. However, caused by the category imbalance of edges and corners, the local ambiguity of edge detection is very serious, which significantly affects the accuracy of predicted outline corners. To tackle this challenge, this article introduces a convolutional RNN embedded with line segment information (LSI-RNN), a novel network that aims to directly detect line segment instead of edges. To achieve this, LSI-RNN utilizes an additional cotraining branch to generate an attraction field map (AFM) by neural discriminative dimensionality reduction (NDDR) layer. Consequently, the conventional classification problem of edges is converted to a regression problem of line segments, thus solving the aforementioned issues. Experimental results over three remote sensing datasets with different spatial resolutions show that the proposed method consistently outperforms other state-of-the-art methods.**

***Index Terms*— Attraction field map (AFM), building outline delineation, convolutional neural network (CNN), recurrent neural network (RNN).**

I. Introduction

THE outlines of buildings extracted from remote sensing images are essential to cartography, urban planning, quick response to natural disasters, and so on. Although the maps of well-established urban areas provide precise building contour, building outlines' information may be outdated or unavailable in a timely manner. In this case, it is motivated to create an automatic and accurate way to delineate the building outlines from new available remote sensing images.

Recently, the artificial intelligence method [1], such as convolutional neural networks (CNNs), has been successfully applied to image classification [2]. The learning of feature extraction and representation through CNN-based deep neural network (DNN) can express high-level features, and hence, DNN shows better performance in building extraction than conventional machine learning methods [3]. The DNN method for delineating building outlines can be technically categorized into two groups, i.e., semantic segmentation methods [4] and instance segmentation methods [5].

Semantic segmentation methods are pixelwise classification using learned features in the given image. A natural semantic segmentation approach is to classify the outline pixels from the background [6]. However, compared with the background pixels, boundary pixels are more difficult to correctly recognize since the number of boundary pixels is rather less and the distribution is more discrete. Therefore, prevailing pixelwise methods first identify building pixels and then utilize additional postprocessing approaches to strengthen building boundaries [7], [8] or use a graph neural network to create the local structure [9]. Although this kind of model could generally achieve good performance statistics [10], they ignore the geometric properties of building boundaries and result in wrong detection, e.g., redundancy in predicted boundary vertexes and inaccuracy of detected shape of the building.

Different from semantic segmentation methods, instance segmentation methods are to allocate label to each individual building. To our knowledge, instance segmentation methods can be broadly divided into three categories.

One kind of instance segmentation method is mask-based models [11]–[14]. They first use a detector to detect buildings and formulate pixelwise classification inside them. Mask region-CNN (Mask R-CNN) [11] is a representative one, which adds a parallel head to the existing detector for pixelwise segmentation. PANet [13] utilizes the bottom-up path augmentation to further improve the accuracy of Mask R-CNN. Mask scoring R-CNN [14] directly optimizes the evaluation metrics to advance the quality of predicted mask. However, this kind of method still originates from pixelwise

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semantic segmentation and has not apt to consider geometric properties of the building shape.

Another kind of instance segmentation approach is contour-based models, and they directly model the boundary of a building instance as an active contour and are first introduced by Kass *et al.* [15] under the name "Snake." Contour-based models first initialize a contour of the object and then move the contour across the energy surface until it halts by minimizing an energy function. Cheng *et al.* [16] used DNN to do this work by minimizing its energy via gradient descent and proposed a deep active ray network (DARNet). Gur *et al.* [17] evolved from it and presented a faster framework with fully differentiable rendering called ACDRNet. Recently, graph convolutional network (GCN) [18] and transformer [19] are utilized to fine-tuning contour-based models. Deep Snake [20] first detects extreme points to generate octagon contour with uniform sampled vertexes, and then, it utilizes GCN to predict the offset of each vertex and deform a new contour fit to the instance. PolyTransform [21] uses mask results as initial contour and adds an additional transformer to predict offsets.

Although contour-based methods achieve attractive efficiency [22], the number of vertexes of the initial contour needed to be manually set to a large number, e.g., 128 in Deep Snake [20], and the predicted contours tend to be rounded shape in corner. This feature makes it unsuitable for the prediction of angular building contours with a small quantity of vertexes in remote sensing images.

The third kind of approach is recurrent neural network (RNN)-based models. Unlike contour-based models, RNN-based models first detect boundary fragments and then find an optimal cycle linking the boundaries into object regions using RNN [23], [24]. Polygon RNN [25] uses CNNs to extract high-level feature maps from a given image and feeds them into RNN to decode the coordinate of per vertex of the instance in order. Polygon RNN++ [26] is an extended version of Polygon RNN, and it adds reinforcement learning [27] and GCN after RNN to improve performance. PolyMapper [28] utilizes an additional CNN-RNN branch to simultaneously delineate road nets and building outlines. Zhao *et al.* [29] improved the structure of CNNs for better feature extraction. More recently, OEC-RNN [30] utilizes richer convolutional features (RCFs) [31] for building edge recognition and integrates it into RNN. Because these models directly predict corner's positions of building boundary, the irregularity of predicted contours in contour-based models can be avoided [29]. Also, this kind of model achieves the best accuracy in delineating building outlines [30].

Generally, the RNN-based model is based on the CNN-RNN architecture. First, it utilizes CNNs to extract high-level feature maps and then feeds them into RNN to decode the coordinate of per corner of the building in order [see Fig. 1(a)]. However, caused by deficiencies of elementary feature extraction and building contour detection by CNNs, the performance of RNN is not always well, especially when the shape of buildings grows more complex [32].

The most recent published work OEC-RNN [30] circumvents this issue by applying an additional edge recognition module to explicitly learn the probability map of edge on buildings and utilizes the edge map as attention to guide RNN to inhibit the prediction sequence of corners deviating the real boundary of buildings. Although OEC-RNN successfully beats the previous state-of-the-art approaches in three open remote sensing datasets, we argue that its attention to edge recognition may lead to local ambiguity for accurate localization due to category imbalance. Two examples are shown in Fig. 2, where the edge recognition map looks like large buffers around the real edges and corners. This ambiguity can confuse RNN, thus failing to predict the position of sharp corners accurately. Moreover, the feature map generation branch and edge recognition branch in OEC-RNN are trained independently, degrading its learning effectiveness.

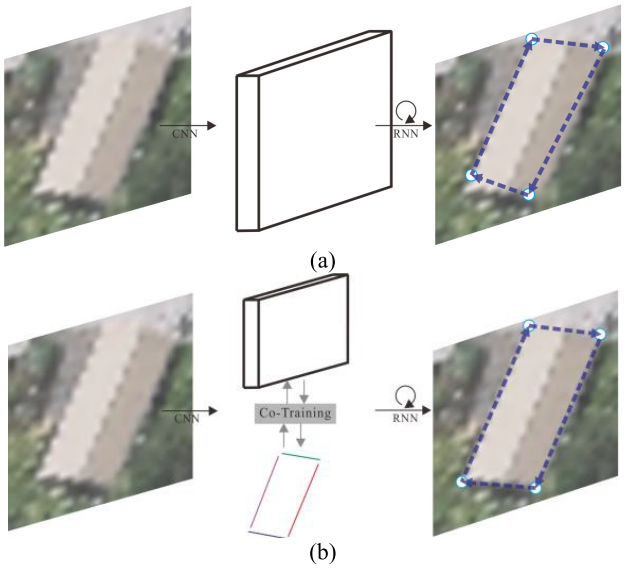

Fig. 1. Comparisons of (a) conventional RNN-based models and (b) LSI-RNN. Our method utilizes additional information of line segments of building to circumvent the deficiencies of building outline detection of CNNs.

In this article, we focus on the local ambiguity of RNN-based models. Our research is motivated by two observations.

1) Line segment contains more geometric information (e.g., direction and distance), which can overcome the local ambiguity inherent to the edge.
2) Recent remarkable progress [33], [34] about line segment detection (LSD) shows that the LSD problem can be posed as a region coloring problem, which promises CNNs to explicitly learn the line segment feature.

The intuitive idea of this article is that we can introduce LSD into the RNN-based model and bridge the detected line segments of building and the corner to be predicted on the building boundary. Specifically, we introduce a multitask building outline delineation algorithm that tackles the aforementioned issues in OEC-RNN by adding an attraction field map (AFM) output, which is a state-of-the-art representation of line segment [33], to a fully convolutional network. While this has an imperceptible effect on training or inference time, not only the AFM guide RNN serially predicts corners of buildings

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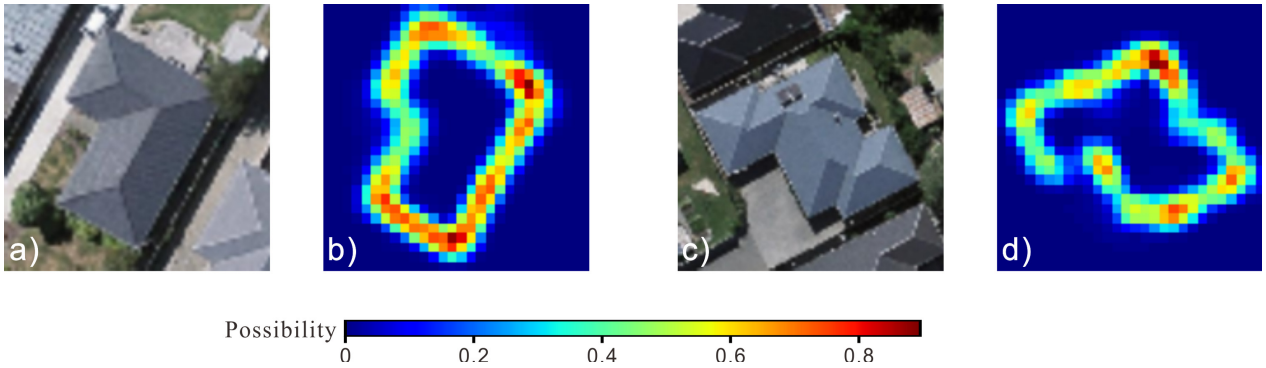

Fig. 2. Two examples of local ambiguity of building edges in OEC-RNN [30]. (b) and (d) Edge detection results of buildings in the example images (a) and (c), respectively.

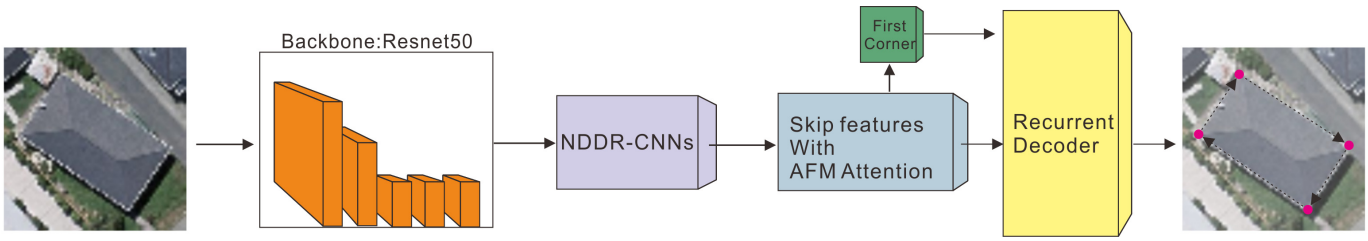

Fig. 3. Workflow of LSI-RNN, which consists of a backbone, NDDR-CNNs, skip features with AFM Attention, and recurrent decoder of corners on a rooftop.

more accurate but also makes the learning procedure of the model more sufficient by the designed cotraining module.

The proposed approach utilizes AFM as additional attention of RNN by the cotraining procedure. AFM uses all the pixels of plane to represent line geometry, unlike the edge map of OEC-RNN that only focuses on the boundary pixels. Since the AFM is a region coloring map, regression loss is utilized as the optimization term of our model, thus solving the category imbalance problem in OEC-RNN. To our knowledge, no existing outline delineation method applies line segments as the prior of RNN. Finally, our model is following multitask architecture, making the learning process more sufficient.

The rest of this article is organized as follows. Section II will introduce the details of our proposed methods. Section III includes our experiments. Section IV describes the discussion. Finally, several conclusions are drawn in Section V.

II. Proposed Methodology

The workflow of RNN embedded with line segment information (LSI-RNN) is shown in Fig. 3, which consists of four sequent components, i.e., backbone, neural discriminative dimensionality reduction (NDDR)-CNNs, skip features with AFM attention, and recurrent decoder. Specifically, the LSI-RNN works as follows.

1) The Resnet-50 [35] is used as a backbone to learning convolutional features from a given image.
2) The convolutional feature is fed into NDDR-CNNs to predict AFM of the building and generate more detailed high-level features, which are named skip features in this article.
3) An attention mechanism is utilized on skip features to get the skip feature fused with AFM. The position of the first corner is predicted from it.
4) Both the first corner' position and the skip feature fused with AFM are sent into RNN for the subsequent prediction of building corners.

A. Backbone

The backbone network performs as a multilevel extractor of the image. We apply the powerful residual network as the backbone. Resnet consists of a series of networks [35], i.e., Resnet-18, Resnet-34, Resnet-50, Resnet-101, and Resnet-152. The number is the hidden layer number in Resnet. The more layer, the better feature extraction ability but more time consuming. Through comprehensive consideration of model accuracy and time consumption, we select Resnet-50 as our backbone (see more details in Section IV-B).

Resnet-50 [35] performs repeated downsampling operations through consecutive layers of CNNs, which results in the low output resolution of the task about building outline delineation. This issue impacts the effectiveness of the model, such as delineating the building with complicated shape. To alleviate it, we reduce the stride of Resnet-50 and introduce dilation factors [36]. This process increases the resolution of each output convolutional feature in Resnet-50 without reducing the receptive field of individual neurons. Before concatenating the output convolutional features, conv filters with a kernel of 3×3, batch normalization [37], ReLU nonlinearities [38], and bilinear upsampling are applied to make the output features from different depths be the same spatial dimensions (i.e., $112 \times 112 \times 64$). The details are shown in Fig. 4.

B. Neural Discriminative Dimensionality Reduction-CNNs

In order to use the AFM of a target building as the attention to guide the sequent prediction of RNN, a simple idea is to apply another branch of CNNs to predict AFM based on

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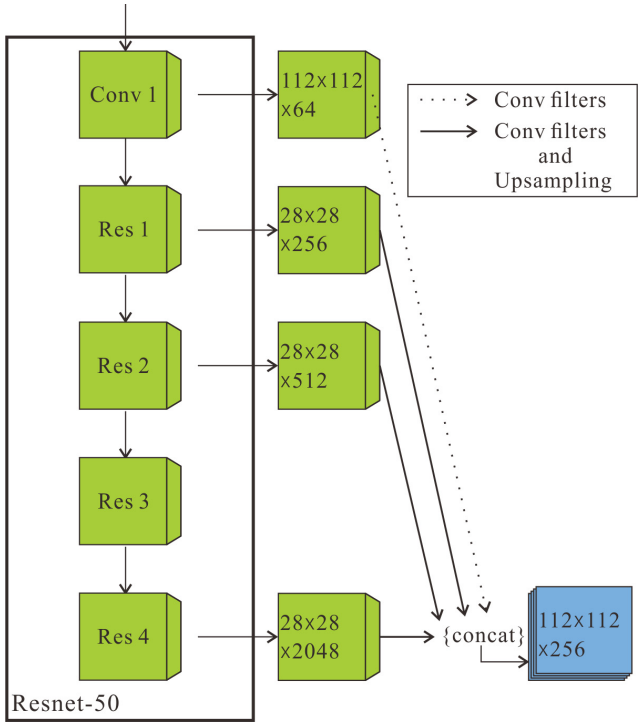

Fig. 4. Processing of Resnet-50 [35]. The number in the cube is the size of tensor, i.e., height × weight × channel. Blue tensor is the final output of Resnet-50.

the extracted high-level convolutional features from the backbone. However, the potential link between these two branches deserves being considered. Therefore, NDDR-CNNs consist of two branches CNNs with multitask learning mechanism.

Following Resnet-50 [35], two structurally identical branches of consecutive convolution and batch normalization layers are utilized to produce skip features ($28 \times 28 \times 126$) and predict the AFM ($28 \times 28 \times 2$). Meanwhile, the skip features and AFM predictions are treated as two different tasks. Because of the potential connection between these two tasks, we refer to the relevant NDDR Layer [39] and introduce it into our model to reinforce the learning process.

NDDR layer is a method for automatically learning the optimal information for layerwise feature fusing in multitask CNNs. Let $F_l^i \in R^{N\times H\times W\times C}$ be the output of an intermediate layer l of task i and there are K tasks. The method first concatenates intermediate features of these tasks in channel dimension

$$F_l = \left[F_l, \ldots, F_l^K\right] \in R^{N\times H\times W\times KC}. \tag{1}$$

F_l is a very high-level tensor that contains redundant information of multiple tasks. Then, a transformation W is learned to reduce the dimensionality of F_l and keep the most discriminative information among different tasks

$$F_l^{i*} = F_l \times W^i. \tag{2}$$

$W^i \in R^{KC\times M}$ is the transformation to be learned for each task i. In this article, M is equal to C, i.e., $F_l^{i*} \in R^{N\times H\times W\times C}$. For optimizing W in the backpropagation of network, the spatial size of $W \in R^{KC\times M}$ is $W \in R^{KC\times 1\times 1\times C}$. Technically, conv filters with a stride of 1 and a kernel size of 1×1 are utilized to represent W.

The structure of NDDR-CNNs is shown in Fig. 5, which consists of two branches, i.e., skip feature branch and AFM branch. The two branches are connected with the NDDR layers. The NDDR layers for these two tasks can be constructed by: 1) concatenating the intermediate features from two branch tasks according to the channel dimension and 2) using the 1×1 convolution, batch normalization layer, and ReLU function to link two tasks.

Intuitively, we use the NDDR layer to make the learning process no longer independent. NDDR layers learn the discriminative information of these two tasks and facilitate the flow of parameters' adjustment in the two branches. Besides, multitask learning approach applies some noise in the training process, which can significantly improve the validation accuracy.

C. AFM Attention Mechanisms

We will first provide the necessary background on AFM (see in Fig.6), which is a key part of our proposed method. Then, we will show the attention mechanisms based on AFM.

Following [33] and [34], AFM is a 2-D field, which assigns two vectors to each point of the plane. AFM is calculated based on the region-partition map. Let I be a crop image containing a building. A line is denoted by $l_i = (x_i^s, x_i^e)$ with two end points, i.e., the start point x_i^s and the endpoint x_i^e. Building outlines are denoted by a set of lines $L = \{l_1, \ldots, l_n\}$.

1) Region-Partition Map: Computing region-partition map for L is a process of assigning every pixel in I to the nearest line segment region. Given a pixel $p \epsilon I$ and a line $l_i \epsilon L$, we first project p onto l_i vertically. If the projection point is not on l_i, the nearest endpoint of l_i is considered as the projection point. Then, the distance between them is calculated as $d(p, l_i)$

$$d(p, l_i; \alpha) = \min\left\|x_i^s + \alpha \cdot \left(x_i^e - x_i^s\right) - p\right\|_2^2 \tag{3}$$

where

$$\alpha_p^* = \arg\min d(p, l_i; \alpha). \tag{4}$$

The projection point is on line segment when $\alpha_p^* \epsilon (0, 1)$ and is on the endpoint when α_p^* is 0 or 1.

The region of l_i is defined as follows, and the region-partition map is denoted as $R = \{R_1, \ldots, R_n\}$:

$$R_i = \{p | p \in I; d(p, l_i) < d(p, l_j), \forall j \neq i, l_j \in L\}. \tag{5}$$

2) Attraction Field Map: Given the region-partition map R_i, corresponding to a line segment, for each pixel p in R_i, the projection point p' on l_i is calculated

$$p' = x_i^s + \alpha_p^* \cdot (x_i^e - x_i^s). \tag{6}$$

For every pixel p, the 2-D attraction or projection vector $(p) = \{x, y\}$, where x and y represent the horizontal and vertical directions of 2-D vector, can be calculated as follows and the AFM is denoted as $A = \{a(p), p \epsilon I\}$:

$$a(p) = p' - p. \tag{7}$$

In LSI-RNN, we utilize a preprocessing module to generate supervised AFM for each building rooftop and apply a branch of CNNs to predict the AFM.

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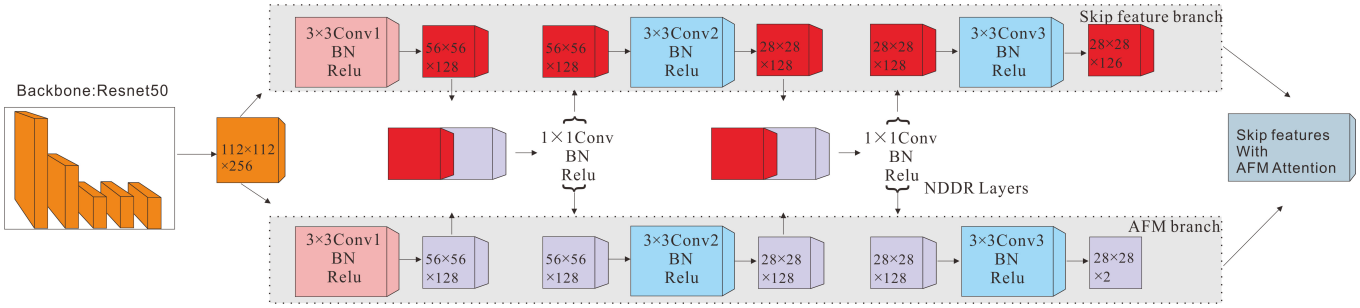


Fig. 5. Structure of NDDR-CNNs.

3) Detected Line Segment as Attention: Traditional edge maps approximate the line segment with only a few pixels in images. Different from that, AFM gives equal attention to all pixels, which contains more information about the line geometry of the building outline. Given the AFM predicting map (F_{afm}) and skip features (F_s) from NDDR-CNNs, we directly concatenate them to get the skip feature fused with AFM ($F_{\text{s-afm}}$).

D. Modified Recurrent Decoder

An RNN is powerful for time-series prediction and can handle complex history information by nonlinear and linear transformation. In this article, we treat corner positions of a building as a sequence and thus utilize RNN to capture the shape of the building outline.

Huang *et al.* [30] used a double-layer Conv-LSTM [40] for outline delineation, which is a widely used RNN variant. We argue that the Conv-LSTM in [40] is still forgetful, and hence, we utilize a mechanism [26] akin to self-attention on it.

The overall structure of our modified Conv-LSTM is shown in Fig. 7. After obtaining the skip features with AFM, we first concatenate it with guidance information. Then, we adopt a self-attention mechanism on it to select the attention to local relevant information. Finally, a double-layer Conv-LSTM is utilized to decode the feature map and delineate the coordinate of corners on the building outline.

In what follows, we will first introduce the workflow of double-layer Conv-LSTM by taking single layer as an example. Then, the self-attention mechanism will be described. At the end of this section, the details of input features will be explained.

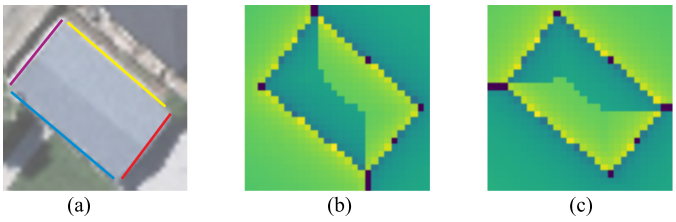
Our Conv-LSTM (single layer) computes its hidden state h_t given the input x'_t according to the following equations:

$$\begin{pmatrix} i_t \\ f_t \\ o_t \\ g_t \end{pmatrix} = W_x \times x'_t + W_h \times h_{t-1} + b \tag{8}$$

$$c_t = \sigma(f_t) \odot c_{t-1} + \sigma(i_t) \odot \tanh(g_t) \tag{9}$$

$$h_t = \sigma(o_t) \odot \tanh(c_t) \tag{10}$$

where the subscript t means this time step, $t-1$ means the previous time step. i, f, and o denote the input, forget, and output gate respectively, σ is the sigmoid function, W_x denotes the convolution kernel of the input, W_h denotes the convolutional kernel of the hidden state, $\odot$ indicates the elementwise product, and C is the cell state.



(a) (b) (c)

Fig. 6. AFM for representation of line segments. (a) Segment of building outline. (b) x component of AFM. (c) y component of AFM.

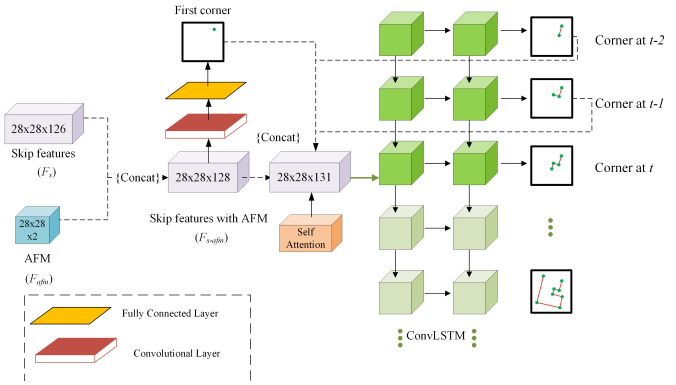


Fig. 7. Structure of modified double-layer Conv-LSTM.

In this article, the kernel size of Conv-LSTM is 3×3. The channel numbers of these two layers are 64 and 16. The input of the final layer is the hidden state of the first layer. The final output of a double-layer Conv-LSTM is the hidden state of the final layer. Note that the final output $\in R^{D\times D\times 16}$, where D is the predicted resolution, we apply a fully connected layer on the output to get the position of the vertex at each time step.

The input x'_t at time step t is calculated through the following mechanisms:

$$\begin{aligned} x'_t &= x_t \odot \alpha_t \\ \alpha_t &= \text{softmax}\big(f_{\text{att}}(x_t, f_1(h_{1,t-1}), f_2(h_{2,t-1}))\big) \end{aligned} \tag{11}$$

where $h_{1,t-1}$ and $h_{2,t-1}$ are the hidden state of the previous time step of two-layer Conv-LSTM. Both f_1 and f_2 are two fully convolutional layers to map $h_{1,t-1}$ and $h_{2,t-1}$ to $R^{D\times D\times 128}$. Here, f_{att} takes the sum of the input and maps it to $R^{D\times D}$ by a fully connected layer. After these processes, a weight matrix for the initial input x_t is obtained and a new input x'_t that focuses only on the relevant information for the next time step can be calculated through elementwise product.

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In this article, the initial input x_t (i.e., the input of ConvLSTM) is a tensor that concatenates multiple features: 1) the skip feature fused with AFM ($F_{\text{s-afm}}$); 2) the position of the first corner; 3) the position of the corner at time step $t-1$; and 4) the position of the corner at time step $t-2$. Since the previous corner position is always given by the last time step, except for the first time. Thus, an additional branch of the network that contains a CNN and a fully connected layer is applied from $F_{\text{s-AFM}}$ to predict the corner probability map. Also, the position of the pixel with the highest probability on this probability map is chosen as the position of the first corner.

III. Experiments and Results

In this section, we begin by introducing the details about the dataset we use for training and validating. Four metrics illustrated in Section III-B are utilized to evaluate the performance of the model. Section III-C includes our strategy of model training and hyperparameters. Section III-D shows the quantitative and qualitative comparisons of the proposed method and other state-of-the-art methods, i.e., Mask R-CNN [11], ACDRNet [17], Polygon RNN [25], Deep Snake [20], and OEC-RNN [30]. Finally, we conduct ablation experiments in Section III-E to test the effectiveness of each component of the proposed method.

A. Datasets

Three datasets with different spatial resolutions are utilized to validate the performance of the proposed method on remote sensing images.

1) *WHU Building Dataset [41]:* This dataset was obtained by manually vectorizing the contours of buildings from the remote sensing images photographed from Christchurch. There are 8189 images with 512 × 512 pixels and three bands (RGB). The spatial resolution is 0.3 m/pixel.
2) *Massachusetts Building Dataset [42]:* This dataset contains 137 remote sensing images with the corresponding labels of size 1500 × 1500, and each image consists of three bands (RGB) at a spatial resolution of 1 m/pixel.
3) *DREAM-A+ Building Dataset [43]:* The remote sensing images of this dataset are from Tianditu of Beijing, Guangzhou, Hangzhou, and Fuzhou, China. The corresponding labels are from Open Street Map (OSM). The images are 224 × 224 with a spatial resolution of 2.5 m/pixel. In this work, we only use three bands (RGB) for a fair comparison. It is worth mentioning that the dataset has a lower spatial resolution, and therefore, the building instances in these data are relatively smaller and the outlines are more difficult to be delineated.

The corresponding label of all datasets are semantic, and we use COCO API[1] to automatically generate supervised corner annotations by the approximate edge from the semantic label and crop images by the smallest enclosed rectangle of the building. In all, there are 136 353 buildings in the WHU Building dataset [41], 101 220 buildings in the Massachusetts Building dataset [42], and 85 783 buildings in the DREAM-A+ Building dataset [43]. The visual view of these datasets is shown in Fig. 8. Also, the proportion of train and validation sets is set to be 8:2.

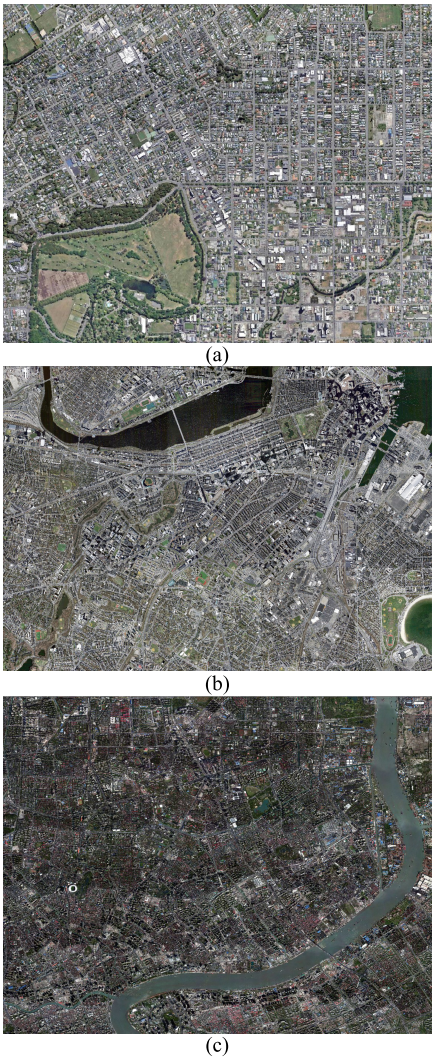

Fig. 8. Example of remote sensing images of three building datasets. (a) WHU Building dataset. (b) Massachusetts Building dataset. (c) DREAM-A+ Building dataset.

B. Evaluation Metrics

Given the predicted polygon and ground-truth outlines of a building, we use four metrics with mean values for evaluation.

Intersection over union (IoU) [44] is the ratio of the predicted and the ground-truth intersection to their union. Assuming that the predicted building polygon composed of predicted corners connected in order is A_p and the ground-truth building polygon is A_g, IoU is calculated as follows:

$$\text{IoU} = \frac{A_g \cap A_p}{A_g \cup A_p}. \tag{12}$$

Rotation error (RE) [30] is a value defined as the deflection angle between the orientations of the predicted polygon and

[1]https://github.com/cocodataset/cocoapi

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the ground truth, which is given as follows:

$$\mathrm{RE} = \left| O(A_g) - O(A_p) \right| \tag{13}$$

where $O(A_g)$ and $O(A_p)$ are the orientations of ground-truth polygon and predicted polygon, respectively.

Hausdorff distance (Hd) [45] is defined as a maximum distance of a set to the nearest point in the other set. It is a maximum function, which can measure the similarity between two sets of points. Suppose that A_p contains a set of corners $A = \{a1, a2, \ldots\}$ and A_g contains a set of corners $B = \{b1, b2, \ldots\}$. Hd is calculated by

$$\mathrm{Hd}(A, B) = \max\left[d_{A,B}, d_{B,A}\right] \tag{14}$$

where

$$d_{A,B} = \max_{a \in A}\left[\min_{b \in B}\|a - b\|\right] \tag{15}$$

$$d_{B,A} = \max_{b \in B}\left[\min_{a \in A}\|b - a\|\right]. \tag{16}$$

Vertex number error (VNE) [30] is a value measuring the difference between the number of predicted vertices and the number of ground-truth vertexes. Given the number of ground-truth vertexes N_g and the predicted vertexes N_p, VNE is calculated by

$$\mathrm{VNE} = \left| N_g - N_p \right|. \tag{17}$$

C. Training Details

We use pretrained Resnet-50[2] as our backbone and train our model by the Adam optimizer with a batch size of 8 and an initial learning rate $\lambda = 1\mathrm{e}^{-3}$. The learning rate was decayed after ten epochs by a factor of 10 and other optimizer parameters are all default values.

1) AFM Predictions: Considering that AFM is numerically very small, which will result in the instability of training process, we apply a reversible nonlinear transformation [33], [34] on supervised information of AFM (a)

$$a' = -\mathrm{sign}\ (a) \times \log(|a| + \varepsilon). \tag{18}$$

Parameter ε is applied to avoid log (0), and we set it $1\mathrm{e}^{-6}$. The loss function we used for predicting AFM is smooth l1 loss [46].

2) Outlines Delineations: The polygon of building outlines we predicted is at resolution $D \times D$, where D is equal to 28. In order to not overpenalize the incorrect predictions closed to the ground-truth corner, we assign nonzero probability mass [25] to those locations that are within a distance of 2. For each time step of RNN, we apply one-hot encoding on the position of every corner and use cross entropy (i.e., cross entropy for 784 categories) as loss. For the task of the prediction of the position of the first corner, we use logistic loss (i.e., cross entropy for two categories) to generate the probability map of each corner and pick the most possible one.

[2]https://download.pytorch.org/models/resnet50-19c8e357.pth

3) Total Loss of LSI-RNN: Huang *et al.* [30] considered that the magnitude of different losses is not the same and the loss value of edge and corner is multiplied by weight to make them the same. Instead, we treat that the loss of line (i.e., loss of AFM prediction) and the loss of polygon are the same in this article, except for the loss of corner. The total loss is given in the following equation:

$$\mathrm{TotalLoss} = \mathrm{AFMLoss} + \mathrm{PolygonLoss} + \varphi\mathrm{CornerLoss} \tag{19}$$

where φ is set to 200 in this article.

D. Comparisons With State-of-the-Art Methods

We use five instance segmentation methods, i.e., Mask R-CNN [11], ACDRNet [17], Polygon RNN [25], Deep Snake [20], and OEC-RNN [30], to compare with our model in terms of both qualitative and quantitative performances.

Mask R-CNN is a two-stage method. It first uses a region proposal network to detect the bounding box of per building and then applies a head network to do pixelwise classification in each bounding box of the given image [11].

ACDRNet is a contour-based model. The method first initializes a circle with diameter of 16 pixels by taking the center of the input image crop. The number of initialized vertices is set to be 32, at which the performance of the model can reach saturation quickly as described in [17].

Polygon RNN [25] is an RNN-based model and structurally similar to LSI-RNN. The architecture of it is CNN-RNN and the backbone is VGG-16 [47]. To make a fair comparison, we replace VGG-16 with Resnet-50 [35], the same as LSI-RNN.

Deep Snake [20] is different from the conventional contour-based methods, and it introduces circular convolution, a kind of GCN, for instance segmentation, thus achieving state-of-the-art performance on the instance segmentation of natural images. The workflow of Deep Snake is followed by: 1) detecting the extreme points of the instance contour of given image; 2) generating an octagon contour with uniformly sampled vertexes based on the extreme points; and 3) iteratively deforming this contour to match the instance boundary.

OEC-RNN [30] is an RNN-based method for building outlines extraction in remote sensing images. Different from other relative methods, it utilizes an additional network named RCF [31] to predict binary edge mask of building and apply the edge recognition result on RNN as attention. The experiments on three remote sensing datasets show that the method achieves the state-of-the-art performance.

The hyperparameters of the above methods adopt those recommended in papers. ACDRNet [17] and Deep Snake [20] are trained for 150 epochs, and the other above models and our proposed method are trained for 15 epochs.

1) Quantitative Evaluation: The quantitative comparison results for different models are listed in Table I. The LSI-RNN obtained the best performance than other methods in three datasets in terms of IoU, RE, Hd, and VNE.

IoU shows the conformity between the predicted polygon and the ground-truth polygon. Our model achieves the highest

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TABLE I
QUANTITATIVE COMPARISONS OF DIFFERENT MODELS. THE BOLD VALUES IN EACH COLUMN MEANS THE BEST ENTRIES

Dataset	Methods	IoU	RE	Hd	VNE
WHU Building	Mask R-CNN	0.8154	0.5055	4.45	7.67
	ACDRNet	0.8016	0.5541	5.12	----
	Polygon RNN	0.8147	0.5543	4.60	2.75
	Deep Snake	0.8404	0.5078	4.38	----
	OEC-RNN	0.8417	0.4974	4.35	2.66
	LSI-RNN	**0.8493**	**0.4878**	**4.22**	**2.44**
Massachusetts Building	Mask R-CNN	0.7482	0.7689	3.52	6.83
	ACDRNet	0.7404	0.7085	2.74	----
	Polygon RNN	0.7473	0.7180	3.11	3.75
	Deep Snake	0.7502	0.7742	3.04	----
	OEC-RNN	0.7755	0.7212	2.36	3.10
	LSI-RNN	**0.7874**	**0.6908**	**2.21**	**3.00**
DREAM-A+ Building	Mask R-CNN	0.7213	0.7621	6.87	8.75
	ACDRNet	0.6988	0.7711	6.69	----
	Polygon RNN	0.7437	0.7193	6.21	6.98
	Deep Snake	0.7413	0.7214	6.11	----
	OEC-RNN	0.7448	0.6980	5.89	6.45
	LSI-RNN	**0.7504**	**0.6742**	**5.55**	**6.24**

metric value, i.e., 0.8493, 0.7874, and 0.7504 in three remote sensing datasets, which outperforms other models. This result demonstrates that our method is more accurate in describing the geometry of the building.

RE shows the difference in the deflection angle of two polygons. Hd is used to measure the spatial difference between two sets of corners and a particular focus on the corner that deviates significantly from the ground truth. Our model gains the lowest values in terms of these two metrics, thus showing the superior control of the model over predicted singularity.

Because contour-based models need to initialize a fixed number of control points, VNE is not considered in ACDRNet and Deep Snake. Since Mask R-CNN directly identifies the pixel of buildings, the number of predicted corners is significantly more than other models, which results in a higher value in VNE. Although the difference among Polygon RNN, OEC-RNN, and LSI-RNN in VNE metrics is not much, LSI-RNN still shows better performance than the others.

In addition, as the spatial resolution of remote sensing images decreases, buildings grow smaller and are more difficult to be delineated. The performance of our model is slightly lower but still outperforms other methods.

2) Qualitative Evaluation: We select three images from each of the three datasets for qualitative comparisons. As shown in Figs. 9 and 10, LSI-RNN can better predict the number and position of outline corners without severe corner deviations, and our model exhibits the best performance with different resolution images as input.

There are obvious differences in the delineating results of outlines among different models in Fig. 9. Although Mask R-CNN can accurately locate the building without the provided bounding box, it can hardly identify the edge of outlines precisely compared with our model. ACDRNet is a traditional contour-based model, which fits the shape of the building by extending the initial polygon composed of control points. As shown in Rows 6 and 7, ACDRNet occasionally leads to under extension. As for Polygon RNN, it often recognizes the number of corners inaccurately. It misses several corners in Rows 3 and 4, and it incorrectly predicts extra isolated corners in Rows 2 and 9. Although the case of under extension is mitigated in Deep Snake, the shape of predicted corners is still rounded, which is not consistent with the regular form of the building. Both OEC-RNN and LSI-RNN show visually similar results compared to the ground truth. However, LSI-RNN is superior to OCE-RNN in terms of object integrity and accuracy compared to others.

In different building datasets, as the resolution decreases, the building instance becomes fuzzier, especially in the boundary and corner of buildings, which poses a big challenge to these methods. For example, Mask R-CNN and ACDRNet may be false to distinguish the building boundary from the background, as shown in Row 7 of Fig. 9. In addition, Polygon RNN along with OEC-RNN often misestimate the position of corners, as shown from Row 4 to the last one. Although the decay of qualitative performance of Deep Snake is lower, it still suffers the bloblike output topology of buildings, which is a serious problem in delineating the building outlines. Although the LSI-RNN could make some errors with the spatial resolution decreasing, compared with all the methods above, LSI-RNN achieved the best visual results.

E. Ablation Experiments

For better understanding why LSI-RNN works well, we conduct ablation studies on the WHU Building dataset [41]. The three combinations are evaluated, including our proposed network, i.e., LSI-RNN, the method of the baseline coupled with only NDDR Layer, i.e., +NDDR, and only AFM attention, i.e., +AFM. All of the experiments are trained for 15 epochs (i.e., 40.5k steps).

1) Baseline: The workflow of our baseline consists of the following structure: 1) a Resnet-50 backbone used in the same way as LSI-RNN; 2) two branches of CNNs in the middle of Fig. 5 with exception of the NDDR layer, i.e., the skip feature branch and AFM branch; 3) the heat map of corners of the building is predicted from skip features by two layers. Note that the skip features do not contain AFM attention; 4) both the predicted position of the first corner and the skip feature are fed into a recurrent decoder, which is the same as that in LSI-RNN.

2) Effectiveness of Each Component: Table II summarizes the evaluation results of ablation experiments. The second row named "+NDDR" only applies the NDDR layer in the AFM prediction branch and skip features generation branch without the concatenation of AFM and skip features. The main difference between it and our baseline is that "+NDDR" modified two branches of CNNs in baseline and extended it to "NDDR-CNNs" described in Section II-B. Compared with LSI-RNN, "+NDDR" does not contain the following

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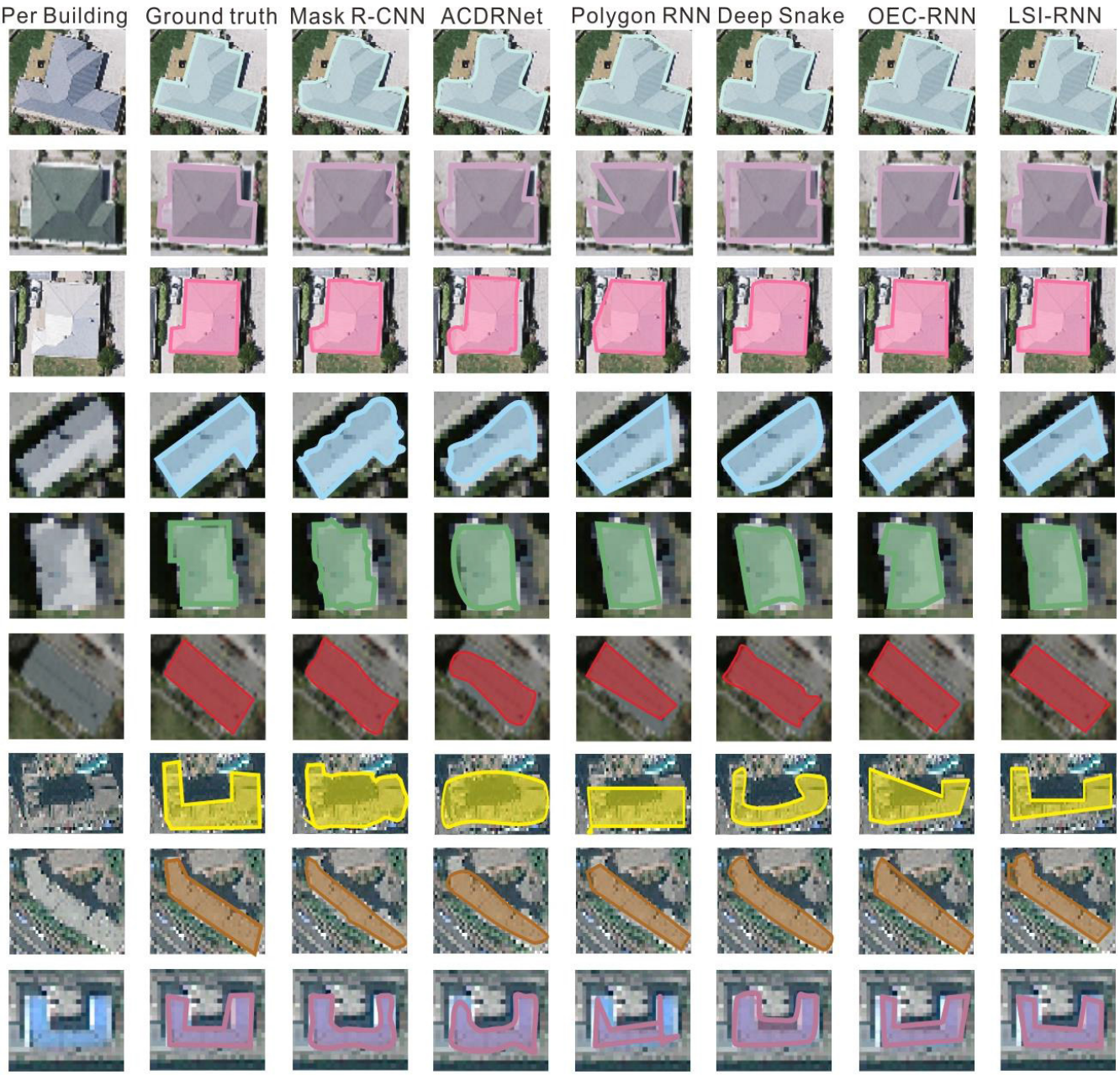

Fig. 9. Qualitative comparisons. These building instances are derived from three datasets. The first to the third rows are from the WHU Building dataset [41], the fourth to the sixth rows are from the Massachusetts Building dataset [42], and the seventh to the ninth rows are from the DREAM-A+ Building dataset [43]. (From Left to Right) First column is the instance images, the second column is the ground truth, and the third column to the eighth column are results of Mask R-CNN [11], ACDRNet [17], Polygon-RNN [25], Deep Snake [20], OEC-RNN [30], and LSI-RNN.

TABLE II
EVALUATION RESULTS OF ABLATION EXPERIMENTS

	IoU	RE	Hd	VNE
Baseline	0.8198	0.5755	4.88	2.67
+NDDR	0.7213	0.6116	5.10	3.18
+AFM attention	0.8288	0.5441	4.75	2.55
LSI-RNN	**0.8493**	**0.4878**	**4.22**	**2.44**

concatenation process. Obviously, its result is worse than our baseline, which will be discussed later.

When exploring the influence of AFM attention, we design an architecture that only concatenates AFM with skip features, as shown in the third row named "+AFM." Compared with our baseline, "+AFM" utilizes two-branch CNNs to generate skip features and predict AFM and concatenate them through channel dimension. Compared with LSI-RNN, "+AFM" removes the NDDR layer connecting the two branches. The comparison of performance between it and our baseline shows a little improvement in terms of four metrics.

Finally, the last row in Table II is the proposed method LSI-RNN, and it combines the NDDR layer and AFM concatenation and yields almost 3% accuracy improvement in terms of IoU.

It is known that whether the position of the first corner predicted from skip features is accurate seriously affects the prediction accuracy of subsequent corners in RNN. As shown

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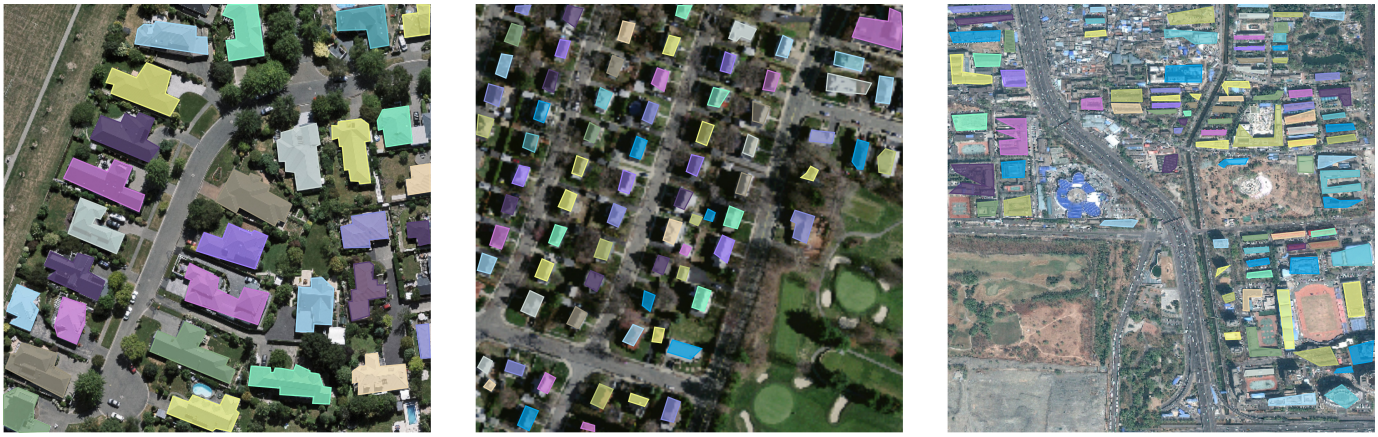

(a) (b) (c)

Fig. 10. Results of LSI-RNN on (a) WHU Building dataset, (b) Massachusetts Building dataset, and (c) DREAM-A+ Building dataset.

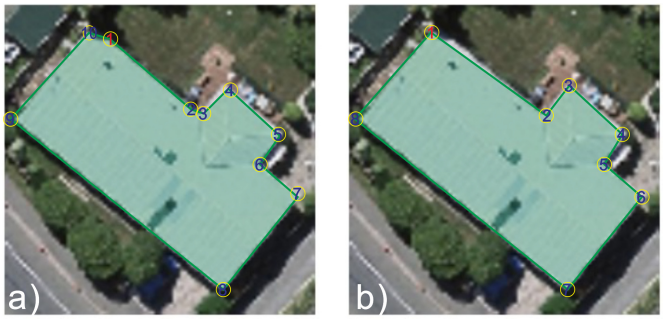

Fig. 11. Comparison of (a) baseline and (b) "+AFM." The number in the yellow circle is the order of sequential prediction of corners. The wrong detection of the first corner can bring a negative effect on the accuracy of the following predicted corners.

TABLE III
PERCENTAGE OF THE FIRST CORNER EXACTLY AT THE GROUND TRUTH

Methods	Baseline	+AFM	+NDDR	LSI-RNN
Percentage	88.5%	**89.6%**	87.8%	**90.0%**

in Table III and Fig. 11, AFM attention can significantly improve the prediction accuracy of the first corner.

Besides, the NDDR layer can improve the generalization ability of the model. As shown in Fig. 12, the training IoU of LSI-RNN is lower than that of "+AFM," but its validation IoU grows higher, which means that the NDDR layer also plays the role of introducing noise. The network only contains an NDDR layer that might be overfitting in the training process, as shown in Fig. 13. Intuitively, without the AFM attention before RNN, the loss of RNN cannot be directly passed into the AFM prediction branch, which results in the insufficient information sharing process brought by the NDDR layer.

Overall, AFM attention is the attention mechanism that explicitly works on RNN. We apply a simple and effective fusion mechanism on skip features to improve the prediction accuracy of the first corner. NDDR layer is more like attention that helps AFM implicitly work on RNN. NDDR layer can incorporate the AFM into the generating branch of skip features in the training process and lead a more clearly implicit direction for RNN. Meanwhile, it also introduces

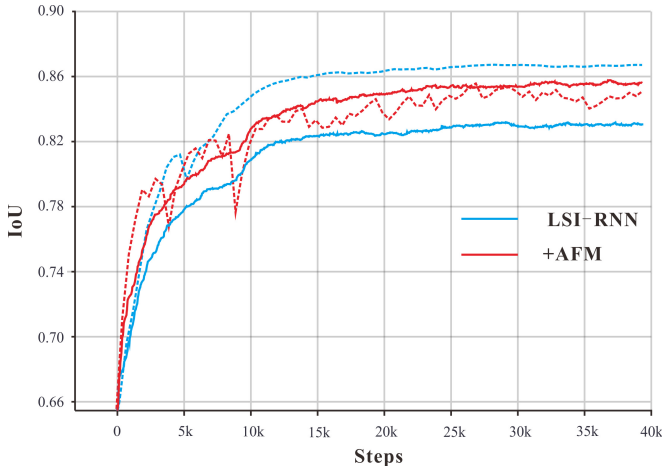

Fig. 12. Comparisons of training and validation IoU between LSI-RNN and "+AFM" in the training process. Blue represents LSI-RNN and red represents "+AFM." Continuous line is training curve and dotted line is validation curve.

noise into network, which improves the generalization ability of the model.

IV. DISCUSSION

A. Parameter Analysis

The global loss function is utilized in the end-to-end training procedure. Because the proposed method is following multitask framework, this function is a sum of losses from three different components, where the hyperparameter φ is the coefficient of the corner loss. In order to investigate the influence of φ with different values on the final result, φ is set as three values, i.e., 100, 200, and 300 to conduct experiments. Note that the WHU Building dataset [41] is selected as an example dataset to carry out the following experiments.

The experiment results of different values of φ are shown in Table IV. It can be seen from Table IV that our method is sensitive to this value. Because of the sparse annotation of corner, the lower value of φ means the less focus of the model on the corner, which will decrease the accuracy dramatically. Furthermore, increasing the value of φ will lead to a slight reduction in terms of evaluation metrics. The best result is obtained when φ is 200, which is also adopted in [30].

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TABLE IV
ACCURACY OF DIFFERENT COEFFICIENTS OF CORNER LOSS (φ)

φ	IoU	RE	Hd	VNE
100	0.8421	0.4894	4.35	2.77
200	**0.8493**	**0.4878**	**4.22**	**2.44**
300	0.8488	0.4882	4.23	2.45

TABLE V
PERFORMANCE OF DIFFERENT BACKBONE ARCHITECTURES

Dataset	Backbone	IoU	RE	Hd	VNE	FPS
WHU Building	Resnet-18	0.7784	0.6784	5.56	3.87	20.8
	Resnet-34	0.8206	0.5211	4.88	3.15	20.5
	Resnet-50	0.8493	0.4878	4.22	2.44	20.1
	Resnet-101	0.8507	0.4814	4.07	2.37	19.4
	Resnet-152	0.8510	0.4808	3.90	2.38	18.5
Massachusetts Building	Resnet-18	0.6973	0.7714	4.18	4.22	24.6
	Resnet-34	0.7588	0.7215	3.75	3.98	24.4
	Resnet-50	0.7874	0.6908	2.21	3.00	24.0
	Resnet-101	0.7887	0.6895	2.16	2.94	23.4
	Resnet-152	0.7901	0.7015	2.24	2.88	22.7
DREAM-A+ Building	Resnet-18	0.6834	0.8094	6.87	7.14	23.5
	Resnet-34	0.7323	0.7571	5.89	7.14	23.3
	Resnet-50	0.7504	0.6742	5.55	6.24	22.9
	Resnet-101	0.7559	0.6684	5.47	6.38	22.2
	Resnet-152	0.7563	0.6713	5.42	6.45	21.4

B. Comparisons of Different Resnet as Backbone

To delve into the impact of different numbers of residual layers in Resnet on our proposed method, we apply five typical Resnet, including Resnet-18, Resnet-34, Resnet-50, Resnet-101, and Resnet-152 as a backbone. The performance of different backbone architectures is reported in Table V, where we report four accuracy evaluation metrics and feature per second (FPS) of input. These experiments are done in Tesla V100 GPU.

It can be seen that as the number of layers in Resnet increases, the accuracy of our model grows higher and the inference speed becomes slower. There is a great gap of accuracy between Resnet-50 and Resnet-34 and a great gap of inference time between Resnet-50 and Resnet-101. Generally speaking, Resnet-50 outperforms other variants of Resnet and is adopted as the backbone of our method based on the comprehensive consideration of both accuracy and inference time.

C. Model Efficiency

The efficiency of different contour-based methods, RNN-based methods, and the proposed method is evaluated in Tesla V100 GPU using the WHU Building Dataset. We reported average inference time and FPS for comparisons in Table VI.

ACDRNet [17] needs to generate a contour surrounding the target building at first, and this initializing process runs on the CPU not the GPU, which causes the slow speed of inference.

TABLE VI
AVERAGE INFERENCE TIME PER OBJECT

	Time (ms)	FPS
ACDRNet	116.2	8.6
Polygon RNN	240.1	4.2
Deep Snake	28.4	35.2
OEC-RNN	60.1	16.7
LSI-RNN	50.0	20.1

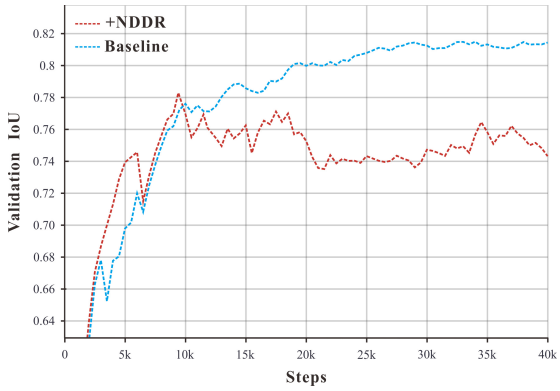

Fig. 13. Validation IoU of baseline and "+NDDR" in the training process. Red is "+NDDR" and blue is our baseline.

In the Polygon RNN [25], all the vertices on the outline of building are used for training, and hence, the model predicts many noncorner points of buildings during the inference phase. Predicting these redundant vertexes dramatically reduces the inference speed. Deep Snake [20] is a modern contour-based architecture and the structure is sample and is friendly to run on the GPU; hence, the speed is fast. However, similar to ACDRNet, Deep Snake needs to propose an initial contour with a fixed number of vertexes (e.g., 128) and then optimize it to the final result. The output building outline is bloblike and does not correspond to the building topology.

OEC-RNN [30] is the previous state-of-the-art method, and it uses an additional complicated RCF [31] branch to bring edge information into the RNN decoder. Compared with the RCF branch in OEC-RNN, the structure of AFM prediction branch along with the NDDR layer in LSI-RNN is rather simple, which brings 3.4 efficiency gains in fps. Although the speed of LSI-RNN is slower than Deep Snake, LSI-RNN shows better quantitative and qualitative results than Deep Snake, as mentioned in Section III-D, and 20.1 frames/s is also acceptable for real-time applications.

D. Limitations

Fig. 14 shows an example of predictions on the curved outline. The designation of LSI-RNN is inclined to delineate the sharp corners on buildings to capture the angular shape of building outline, which causes the neglect of the smooth vertexes on the curve, as shown in Fig. 14(b). In the contrast, the contour-based model exploits redundant initialization corners to fit the outline of buildings. As shown in Fig. 14(c), though the result seems bloblike on sharp corners, the contour-based model can better handle the curve shape of buildings. Nonetheless, most of the buildings are with angular shape, as described

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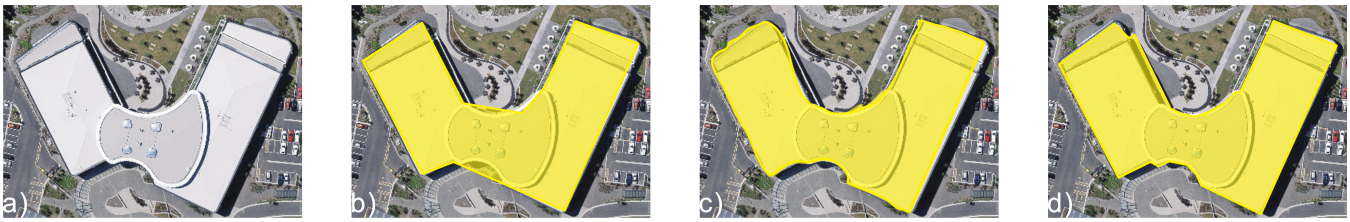

Fig. 14. Example of predictions on curved outline. (a) Image. (b) Results of LSI-RNN. (c) Results of Deep Snake. (d) Ground truth.

in Section III-D, and our method has reached a level adequate for coping with building outline delineation.

V. Conclusion

In this article, we introduce LSI-RNN, an RNN-based model, for accurately and automatically delineating the outline of buildings in remote sensing images. The overall framework of our model is based on the CNN-RNN architecture. CNNs are utilized for feature extracting from the given image and RNN is served as a decoder to predict the position of per corner of the building at a time.

Following the previous research about building outline delineation, we introduce several changes on the previous work [25], [26], [30], including replacing the oversmooth binary edge recognition of the target building with attraction field representation and applying the NDDR layer to enhance the information sharing mechanism between CNNs. Apart from the metrics of precision such as IoU, we also utilize RE, Hd, and VNE to measure the control over predicted singularity of the model.

Ablation experiments of LSI-RNN demonstrate that AFM attention can significantly improve the prediction accuracy of the first corner and the NDDR layer can implicitly incorporate AFM into the RNN, which leads to a more clearly circulation direction for RNN and promote the generalization ability of the model. Comparison experiments over three remote sensing datasets show that LSI-RNN outperforms other state-of-the-art methods in terms of both quantitative and qualitative evaluation methods.

In future work, we will try to investigate whether line-guided information does apply to other segmentation models.

References

[1] J. Wei *et al.*, "Cloud detection for Landsat imagery by combining the random forest and superpixels extracted via energy-driven sampling segmentation approaches," *Remote Sens. Environ.*, vol. 248, Oct. 2020, Art. no. 112005.

[2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, pp. 1097–1105.

[3] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[4] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[5] T.-Y. Lin *et al.*, "Microsoft coco: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2014, pp. 740–755.

[6] D. S. Lee, J. Shan, and J. S. Bethel, "Class-guided building extraction from Ikonos imagery," *Photogrammetric Eng. Remote Sens.*, vol. 69, no. 2, pp. 143–150, Feb. 2003.

[7] K. Zhao, J. Kang, J. Jung, and G. Sohn, "Building extraction from satellite images using mask R-CNN with building boundary regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 242–246.

[8] S. Wei, S. Ji, and M. Lu, "Toward automatic building footprint delineation from aerial images using CNN and regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2178–2189, Mar. 2020.

[9] Y. Shi, Q. Li, and X. X. Zhu, "Building segmentation through a gated graph convolutional neural network with deep structured feature embedding," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 184–197, Jan. 2020.

[10] N. Yang and H. Tang, "GeoBoost: An incremental deep learning approach toward global mapping of buildings from VHR remote sensing images," *Remote Sens.*, vol. 12, no. 11, p. 1794, Jun. 2020.

[11] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," in *Proc. ICCV*, Oct. 2017, pp. 2961–2969.

[12] C. Ji and H. Tang, "Number of building stories estimation from monocular satellite image using a modified mask R-CNN," *Remote Sens.*, vol. 12, no. 22, p. 3833, Nov. 2020.

[13] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9197–9206.

[14] Z. Huang, L. Huang, Y. Gong, C. Huang, and X. Wang, "Mask scoring R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6409–6418.

[15] M. Kass, A. Witkin, and D. Terzopoulos, "Snakes: Active contour models," *Int. J. Comput. Vis.*, vol. 1, no. 4, pp. 321–331, 1988.

[16] D. Cheng, R. Liao, S. Fidler, and R. Urtasun, "DARNet: Deep active ray network for building segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7431–7439.

[17] S. Gur, T. Shaharabany, and L. Wolf, "End to end trainable active contours via differentiable rendering," 2019, *arXiv:1912.00367*.

[18] F. Scarselli *et al.*, "The graph neural network," *IEEE Trans. Neural Networks.*, vol. 20, no. 1, pp. 61–80, Mar. 2009.

[19] A. Vaswani *et al.*, "Attention is all you need," 2017, *arXiv:1706.03762*.

[20] S. Peng, W. Jiang, H. Pi, X. Li, H. Bao, and X. Zhou, "Deep snake for real-time instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8533–8542.

[21] J. Liang, N. Homayounfar, W.-C. Ma, Y. Xiong, R. Hu, and R. Urtasun, "PolyTransform: Deep polygon transformer for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9131–9140.

[22] T. H. N. Le *et al.*, "Active contour model in deep learning era: A revise and review," in *Studies in Computational Intelligence*, vol. 890. Cham, Switzerland: Springer, Mar. 2020, pp. 231–260.

[23] J. Wang, Y. Yang, J. Mao, Z. Huang, C. Huang, and W. Xu, "CNN-RNN: A unified framework for multi-label image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2285–2294.

[24] W. Huang, Z. Liu, H. Tang, and J. Ge, "Sequentially delineation of rooftops with holes from VHR aerial images using a convolutional recurrent neural network," *Remote Sens.*, vol. 13, no. 21, p. 4271, Oct. 2021.

[25] L. Castrejon, K. Kundu, R. Urtasun, and S. Fidler, "Annotating object instances with a polygon-RNN," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5230–5238.

[26] D. Acuna, H. Ling, A. Kar, and S. Fidler, "Efficient interactive annotation of segmentation datasets with Polygon-RNN++," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 859–868.

[27] C. Szepesvári, "Algorithms for reinforcement learning," *Synth. Lectures Artif. Intell. Mach. Learn.*, vol. 4, no. 1, pp. 1–103, Jun. 2009.

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[28] Z. Li, J. D. Wegner, and A. Lucchi, "Topological map extraction from overhead images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1715–1724.
[29] W. Zhao, C. Persello, and A. Stein, "Building outline delineation: From aerial images to polygons with an improved end-to-end learning framework," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 119–131, May 2021.
[30] W. Huang, H. Tang, and P. Xu, "OEC-RNN: Object-oriented delineation of rooftops with edges and corners using the recurrent neural network from the aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, doi: 10.1109/TGRS.2021.3076098.
[31] Y. Liu, M.-M. Cheng, X. Hu, K. Wang, and X. Bai, "Richer convolutional features for edge detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3000–3009.
[32] Y. Zhu, B. Huang, J. Gao, E. Huang, and H. Chen, "Adaptive polygon generation algorithm for automatic building extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3081582.
[33] N. Xue, S. Bai, F. Wang, G.-S. Xia, T. Wu, and L. Zhang, "Learning attraction field representation for robust line segment detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1595–1603.
[34] N. Xue *et al.*, "Holistically-attracted wireframe parsing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2788–2797.
[35] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[36] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.
[37] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. 32nd Int. Conf. Mach. Learn.*, Lille, France, vol. 2015, vol. 37, pp. 448–456.
[38] V. Nair and G. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 807–814.
[39] Y. Gao, J. Ma, M. Zhao, W. Liu, and A. L. Yuille, "NDDR-CNN: Layerwise feature fusing in multi-task CNNs by neural discriminative dimensionality reduction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3205–3214.
[40] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," 2015, *arXiv:1506.04214*.
[41] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.
[42] M. Volodymyr, "Machine learning for aerial image labeling," Ph.D. dissertation, Dept. Comput. Sci., Univ. Toronto, Toronto, ON, Canada, 2013.
[43] T. Zhang, H. Tang, Y. Ding, P. Li, C. Ji, and P. Xu, "FSRSS-Net: high-resolution mapping of buildings from middle-resolution satellite images using a super-resolution semantic segmentation network," *Remote Sens.*, vol. 13, no. 12, p. 2290, Jun. 2021.
[44] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Sep. 2009.
[45] J. Avbelj, R. M´uller, and R. Bamler, "A metric for polygon comparison and building extraction evaluation," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 170–174, Jan. 2015.
[46] Y. Liu and L. Jin, "Deep matching prior network: Toward tighter multi-oriented text detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1962–1969.
[47] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

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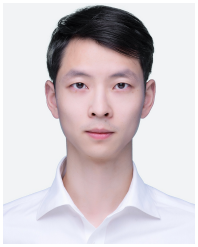

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